

CUSTOMER CHURN ANALYSIS

End-to-End Data Analytics Project Report

Python | SQL (MySQL) | Power BI | Flask

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Telco Customer Churn Dataset — 7,032 records

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1. Introduction

Customer churn analysis helps organizations understand why customers discontinue services. This project analyzes telecom customer data using Python, SQL, and Power BI to identify churn patterns, and extends the analysis into a deployed Flask web application that scores churn risk for a customer in real time.

2. Problem Statement

Telecommunication companies frequently experience customer attrition driven by factors such as pricing, contract type, internet service, and payment method. High churn rates directly reduce revenue and slow customer growth. This project identifies the key drivers of churn and supports data-driven retention strategy.

3. Project Objectives

- Analyze customer churn behavior across the full customer base.
- Identify the factors most strongly associated with churn.
- Perform SQL-based business analysis for stakeholder reporting.
- Build an interactive Power BI dashboard for ongoing monitoring.
- Train and deploy a machine learning model for real-time churn scoring.

4. Methodology

The project followed a structured workflow: data collection, data cleaning, exploratory data analysis, SQL business queries, dashboard development, model training, and deployment. Python (Pandas, Matplotlib, Seaborn, scikit-learn) was used for analysis and modeling, MySQL for business queries, Power BI for dashboarding, and Flask for serving the trained model as a web application.

5. Dataset Description

The dataset contains telecom customer information across 7,032 records and 21 columns, including:

- Customer demographics (gender, senior citizen status, partner, dependents)
- Account information (tenure, contract type, paperless billing, payment method)
- Subscribed services (phone, internet, online security, tech support, streaming)
- Billing (monthly charges, total charges)
- Churn status (Yes / No)

Source: *Telco Customer Churn* dataset, Kaggle (blastchar/telco-customer-churn).

6. Data Cleaning Process

Raw data rarely arrives analysis-ready. The following preprocessing steps were performed in Python/Pandas before any analysis:

- Converted TotalCharges from text to numeric, coercing the small number of blank entries (new customers with zero tenure) to missing values.
- Verified and handled missing values across all columns.
- Checked for and removed duplicate customer records.
- Corrected column data types ahead of aggregation and modeling.
- Exported the cleaned dataset for downstream SQL and Power BI use.

Data cleaning improved data quality and ensured the KPI and model numbers reported throughout this document are reliable.

7. Exploratory Data Analysis

EDA was performed using Matplotlib and Seaborn to surface churn trends ahead of SQL and dashboard work. Two of the core charts are reproduced below directly from the analysis notebook.

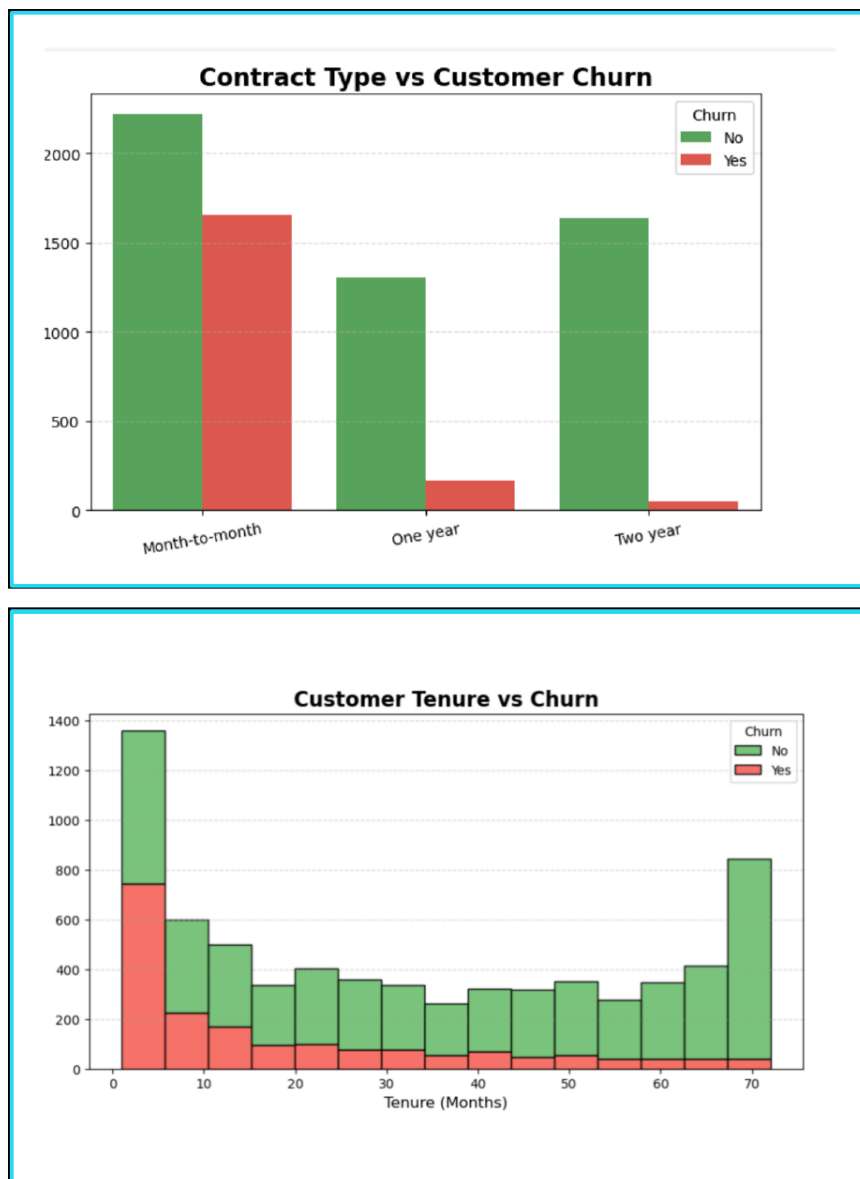


Figure 1. Contract type vs. churn (top) and customer tenure vs. churn (bottom), generated directly from *churn_analysis.ipynb*.

Analyses performed

- Customer churn distribution
- Contract type vs. churn
- Payment method vs. churn
- Monthly charges vs. churn
- Customer tenure vs. churn

8. SQL Analysis

Five business queries were executed in MySQL Workbench against the cleaned dataset loaded into a `churn_data` table. Each query and its purpose:

#	Query	Business purpose
1	Total Customers	Establishes the analysis denominator (7,032).
2	Churned Customers	Counts customers with Churn = 'Yes' (1,869).
3	Churn by Contract Type	Ranks contract types by churn volume.
4	Churn by Payment Method	Ranks payment methods by churn volume.
5	Avg. Monthly Charges by Churn	Compares spend between churned and retained customers.

All five queries were re-verified against a test database during the most recent project review, confirming correct syntax and expected result shape (see *Churn_Analysis_sql.sql*).

9. Power BI Dashboard

An interactive Power BI dashboard was built to give a business-friendly, filterable view of churn behavior. It includes:

- 4 KPI cards: Total Customers, Active (Retained) Customers, Churned Customers, and Churn Rate (%)
- 3 analysis charts: Contract Type vs. Churn, Average Monthly Charges by Churn, and Payment Method vs. Churn
- A Contract-type slicer for interactive filtering

Note: an earlier export of this dashboard had its Contract slicer left filtered to "Month-to-month" only, which understated the KPI totals. The slicer's persisted filter state has since been corrected in the project's `.pbix` file so the dashboard loads against the full, unfiltered dataset.

10. KPI Metrics

KPI	Value
Total Customers	7,032
Retained Customers	5,163
Churned Customers	1,869
Churn Rate	26.58%
Retention Rate	73.42%
Average Monthly Charges	Rs. 64.80
Model Accuracy	78.32%

11. Machine Learning Model & Deployment

Beyond historical analysis, this project trains a classifier to score churn risk for an individual customer, and serves it through a web application for real-time use.

Model

- Algorithm: Random Forest Classifier (scikit-learn)
- Features: 19 customer attributes (categorical fields label-encoded)
- Train/test split: 80/20
- Class balancing: `class_weight='balanced'`, to handle the ~27%/73% churn imbalance
- Accuracy: 78.32% on the held-out test set

Deployment

The trained model, its label encoders, and the exact training-time feature order are serialized with `pickle` and loaded by a Flask application (`app.py`). A form-based frontend (`templates/index.html`) collects the 19 inputs and posts them to a `/predict` endpoint, which returns a churn probability, a retain probability, and a categorical risk level (Low / Medium / High Risk) with an associated color code for the UI.

This end-to-end path — trained model in, web form in, live probability out — was tested directly: the Flask app was exercised with a test client against both the homepage and the prediction endpoint, confirming a working request/response cycle end to end, including graceful handling of a category value outside the training set.

12. Key Findings

- Month-to-month contract customers have the highest churn rate of the three contract types.
- Customers paying by electronic check churn more than those on other payment methods.
- Higher monthly charges are associated with increased churn probability.
- Long-term (one- and two-year) contracts substantially improve retention.
- Churn is concentrated in the first 12 months of tenure — new customers are the highest-risk segment.

13. Challenges Faced & Resolutions

Two categories of challenge came up over the course of this project: ones encountered during the original analysis, and a second round found during a later code/data review.

During analysis

- Handling missing values in TotalCharges
- Importing CSV data into MySQL for SQL analysis
- Power BI visual alignment and formatting

During review

- The training notebook saved the model in a cell that executed before the cell that actually trained it, which would fail on a clean top-to-bottom run; cell order was corrected and re-verified by executing the full notebook end to end.
- Relative file paths in the notebook pointed one directory above the project after the notebook was relocated into the repository; paths were corrected and re-tested.
- The Power BI dashboard's Contract slicer had a persisted filter selecting only "Month-to-month", understating every KPI card; the filter was removed directly from the report's saved state.

14. Business Recommendations

- Promote long-term contracts through discount incentives.
- Improve the experience for electronic-check payment users.
- Proactively monitor customers with high monthly charges.
- Introduce loyalty programs targeted at customers in their first 12 months.
- Bundle tech support and online security with base plans to reduce churn.

15. Conclusion

This project demonstrates an end-to-end customer churn analytics pipeline: cleaning and exploring telecom customer data in Python, answering business questions in SQL, visualizing churn behavior in Power BI, and deploying a trained classifier as a live, usable web tool. Beyond the original analysis, a subsequent review and fix pass verified the notebook executes cleanly start to finish, the SQL queries run correctly, the Flask application serves real predictions, and the Power BI dashboard reflects the full, unfiltered dataset — leaving the project in a reproducible, portfolio-ready state.

16. References

- Telco Customer Churn Dataset — Kaggle ([blastchar/telco-customer-churn](#))
- Python: Pandas, NumPy, Matplotlib, Seaborn, scikit-learn documentation
- MySQL Workbench documentation
- Microsoft Power BI documentation
- Flask documentation